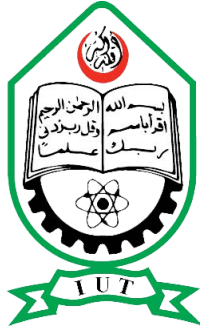


ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)



Project Report

SWE 4304: Software Project Lab I

Project Title

StudyPet - Your Virtual Study Companion

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Project Overview

StudyPet is a Command-Line Interface (CLI) based Study Companion Pet system designed to help students maintain consistent study habits through gamification and emotional engagement. The system allows users to create an account, set daily study goals, and interact with a virtual pet that reflects the user's study behavior, mood, and consistency.

Problem Statement

Many students struggle to maintain consistent study habits, stay motivated, and manage their time effectively. Traditional productivity tools often focus only on task tracking and lack emotional engagement, feedback on long-term study patterns, and support for mental well-being. As a result, students may lose motivation, procrastinate, or experience burnout due to unbalanced study routines.

Potential Solution

StudyPet proposes a gamified study assistant where a virtual pet reflects the user's study behavior, mood, and consistency. By combining Pomodoro-based study sessions, a reward system using coins, mood tracking, and visual analytics such as a study heatmap and weekly reports, the system encourages discipline, engagement, and self-reflection. The interactive pet, along with rewards, penalties, and stress-relief activities, helps students build healthier and more consistent study habits.

Project Structure Tree

StudyPet

```
|— .gitignore
|— main.py
|— README.md
|— src/
    |— __init__.py
    |— core/          # Core business logic
        |— __init__.py
        |— analytics.py    # Analytics and performance calculations
        |— shop.py        # Shop and item-related logic
        |— wallet.py      # Wallet, balance, and transactions
    |— custom/        # Custom utility functions
        |— __init__.py
        |— custom_hash.py  # Hashing utilities
        |— custom_input.py # Input handling helpers
        |— custom_random.py # Randomization helpers
        |— custom_text.py  # Text formatting utilities
        |— custom_validation.py # Input/data validation
    |— interface/     # User interface layer
        |— __init__.py
        |— ui.py       # Main UI components
    |— pet/           # Virtual pet module
        |— __init__.py
        |— animation.py  # Pet animation handling
        |— evolution.py  # Pet growth/evolution logic
        |— pet.py        # Pet model and actions
    |— study/         # Study and learning module
        |— __init__.py
```

```

|— reflection.py      # Reflection generation/processing
|— study.py          # Main study workflow
|— user_reflection.py # User reflection management
|— weekly_report.py  # Weekly study report generation
|
|— quiz/             # Quiz subsystem
|   |— __init__.py
|   |— quiz.py
|   |— quiz_analytics.py
|   |— quiz_charts.py
|   |— quiz_config_helpers.py
|   |— quiz_dashboard.py
|   |— quiz_file_io.py
|   |— quiz_goal.py
|   |— quiz_goal_helpers.py
|   |— quiz_syllabus.py
|   |— quiz_text_helpers.py
|   |— quiz_trend.py
|   |— quiz_ui_boxes.py
|   |— quiz_ui_input.py
|   |— quiz_ui_marks.py
|
|— study_planner/    # Study planner subsystem
|   |— __init__.py
|   |— study_planner.py
|   |— study_planner_config_helpers.py
|   |— study_planner_file_io.py
|   |— study_planner_plan.py
|   |— study_planner_profile.py
|   |— study_planner_progress.py
|   |— study_planner_recovery.py
|   |— study_planner_text_helpers.py
|   |— study_planner_ui_boxes.py
|   |— study_planner_ui_input.py
|   |— study_planner_ui_subjects.py
|
|— system/           # System-level services

```

```

|   |— __init__.py
|   |— auth.py      # Authentication and access control
|   |— navigation.py # Navigation between modules/pages
|   |— storage.py    # Data storage and persistence
|
|— wellbeing/        # Wellbeing and recreation module
   |— __init__.py
   |— wellbeing.py   # Wellbeing-related features
   |— recreation.py  # Recreation and engagement activities

```

Implemented Features

1. 🗝️ Authentication & user data

1. Registration

- Collects email, nickname, password, and pet theme
- Password input is masked and validated with strength rules (≥ 8 characters, ≥ 1 uppercase, ≥ 1 digit)
- Validates email format before account creation
- Initializes default user profile values (health, energy, coins, streak fields, inventory, mood, theme, animation, last login, etc.)
- Stores passwords as salted hashes using the custom hashing module

2. Login

- Validates user credentials (email and password)
- Applies inactivity penalty after ≥ 7 days
- Updates last login timestamp
- Returns user state for dashboard

3. Settings & account management

- Supports updates for name, email, password, pet, theme, and animation
- Supports account deletion
- Handles migration of user-keyed data when email is changed

2. User interface utilities

- Provides console helpers (clear screen, pause, menu selection)
- Implements styled UI boxes and branded splash/header screens
- Displays user summaries, stats views, and navigation menus
- Includes mood picker, analytics menu, reflection menu
- Supports theme and animation selection
- Implements global shortcuts for back and exit

3. Virtual pet

- Enforces pet defaults via centralized helper
- Implements health and coin adjustment utilities
- Displays status with ASCII visuals, level/stage, and inventory
- Supports pet abilities (Cat, Dog, Bunny)
- Uses shared evolution-stage helper across modules

4. Shop and feeding

- Provides normal and premium food purchase flow
- Feeding consumes inventory and restores health (varies by food type)
- Integrates wallet and transaction history

5. Study sessions

- Supports Pomodoro sessions (preset and custom durations)
- Integrates planner-derived topics, difficulty, and timing
- Displays animated countdown with pet visuals
- Adjusts rewards and penalties based on difficulty
- Includes consecutive-session bonus mode
- Cancels sessions on pause-budget overrun
- Generates session logs for analytics and persistence

- Includes DEV mode for testing

6. 🎬 Animation system

- Provides ASCII animation sets for all pets (mood and stage variants)
- Implements frame cycling and countdown rendering
- Includes evolution animation sequences

7. 🧬 Evolution

- Uses level-based stage thresholds
- Applies study-hour milestones and tired-streak boosts
- Returns evolution status for UI feedback

8. ❤️ Well-being and mood tracking

- Stores mood logs persistently
- Prevents duplicate same-day entries
- Computes tired streak from mood history
- Applies tired penalties and burnout handling
- Manages energy updates and recovery

9. 🎮 Recreation

- Unlocks features based on tired streak or daily activity
- Provides guided meditation routines
- Includes mini-game functionality
- Uses study logs for daily session tracking

10. 📊 Study analytics

- Provides heatmap-style analytics over selectable ranges
- Tracks total minutes, sessions, current streak, longest streak, best day
- Uses menu-driven console rendering

11. 📅 Weekly report

- Generates last-7-day summaries (study time, consistency, mood trends, pet state)
- Computes productivity score
- Provides improvement suggestions
- Supports report snapshot saving

12. 🧠 Academic performance tracker (Quiz)

- Supports quiz and mid-term marks tracking
- Tracks syllabus and personal coverage
- Provides trend analysis and performance insights
- Generates text-based charts
- Displays per-subject and overall dashboards
- Uses dedicated storage with legacy compatibility
- Includes full-featured quiz management menu

13. 💾 Storage helpers

- Implements safe read/write operations
- Auto-creates required directories
- Handles JSON corruption scenarios
- Includes backup and recovery for study logs

14. 🧩 Utility functions

- Provides shared utilities (validation, hashing, randomness, text formatting, timed input)
- Includes specialized helpers for quiz and analytics processing

15. Entry points and orchestration

- Manages register/login flow and dashboard routing
- Executes post-login checks (mood, penalties, recreation unlocks)
- Provides full dashboard navigation (study, pet, shop, stats, mood, quiz, analytics, reports, planner, reflection, settings, wallet, logout)
- Updates user state, logs sessions, and triggers evolution/recreation

Tools & Technologies

- ❖ **Programming Language :** Python
- ❖ **User Interface:** Command-Line Interface (CLI)
- ❖ **Development Environment:** Visual Studio Code (VS Code)
- ❖ **Data Management:** Python file handling for persistent data storage
- ❖ **Version Control:** Git
- ❖ **Repository Hosting Platform:** GitHub

Application Domain

This project falls within the **Education Technology (EdTech)** and **student productivity** domain. It is designed to motivate students, track study progress, and provide feedback to support academic discipline and routine building. Our

virtual pet system makes learning more engaging and effective by gamifying the whole study experience.

Project Timeline

Week 1 — Requirement analysis & project planning

Week 2 — System design, architecture & flow diagram

Week 3 — Core framework & basic feature setup

Week 4 — Account creation & user profile system

Week 5 — Pet core mechanics (health, energy, mood)

Week 6 — Topic-based study & timer system

Week 7 — Feeding system & reward logic

Week 8 — Pet shop features & animations

Week 9 — Mood tracking & health logic

Week 10 — Login streaks & goal tracking along with testing

Week 11 — Heatmap & study activity tracking along with testing

Week 12 — Reports & quiz/feedback input along with testing

Week 13 — Prediction models & insights along with testing

Contribution

Student ID	Student Name	Contribution
230042120	Safia Aman Anika	1. storage.py 2. auth.py 3. animation.py 4. evolution.py 5. weekly_report.py
230042124	Tasnia Farzana	1. study.py 2. quiz.py 3. studyplanner.py
230042126	Mushfeka Ferdousi	1. main.py (Merged all the modules of all branches and integrated them) 2. storage.py 3. navigation.py 4. ui.py 5. recreation.py 6. wellbeing.py
230042154	Aksan Anan Ria	1. pet.py 2. shop.py 3. analytics.py 4. reflection.py 5. wallet.py

Future Work

The project can be extended to enhance accessibility, engagement, and personalization for students:

- ❖ **Website and Mobile Versions:** Build responsive web and mobile apps with improved, engaging interfaces.
- ❖ **Cloud Sync & Multi-Device Support:** Enable seamless access and data synchronization across multiple devices.
- ❖ **AI-Based Study Recommendations:** Provide personalized study suggestions based on user behavior and performance.
- ❖ **Social & Collaborative Features:** Facilitate group learning, interaction, and peer motivation.
- ❖ **Advanced Analytics & Reports:** Offer insights into study habits, productivity trends, and overall progress.

Source Repository Link

<https://github.com/zenken24/StudyPet>

Conclusion

The StudyPet system provides an engaging and structured approach to studying by blending productivity techniques with emotional and reward-based feedback. The project emphasizes consistency, self-awareness, and motivation, making it a practical and interactive tool for students.